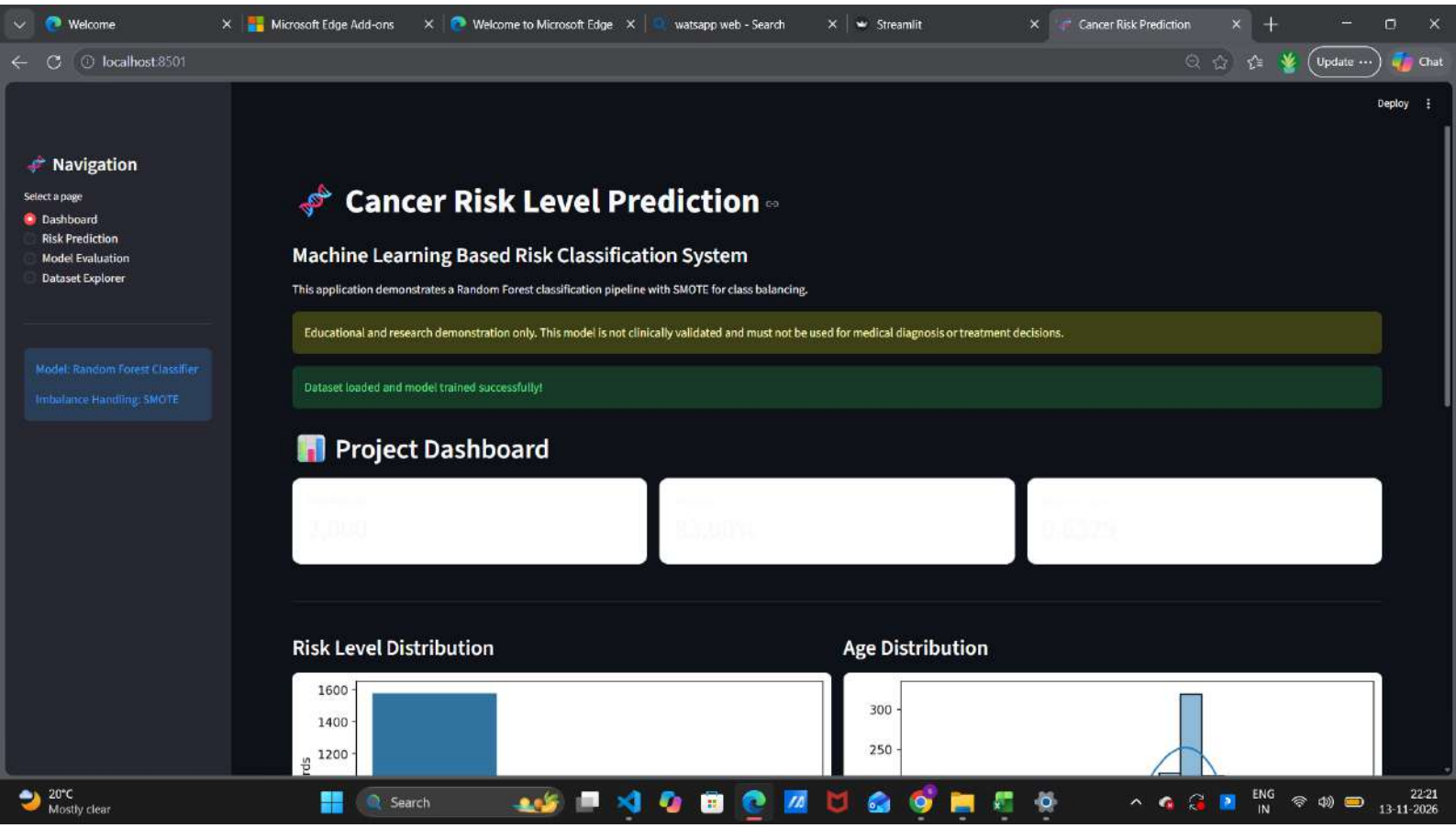

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Dataset Explorer
Model: Random Forest Classifier
Imbalance Handling: SMOTE
Cancer Risk Level Prediction
Machine Learning Based Risk Classification System
This application demonstrates a Random Forest classification pipeline with SMOTE for class balancing.
Educational and research demonstration only. This model is not clinically validated and must not be used for medical diagnosis or treatment decisions.
Dataset loaded and model trained successfully!
Project Dashboard
Risk Level Distribution
Age Distribution
1600
1400
1200
300
250
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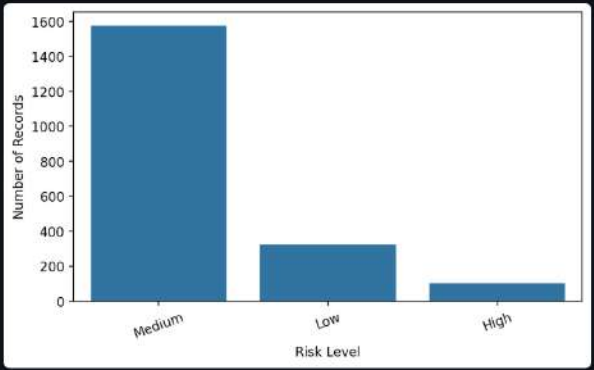
Select a page

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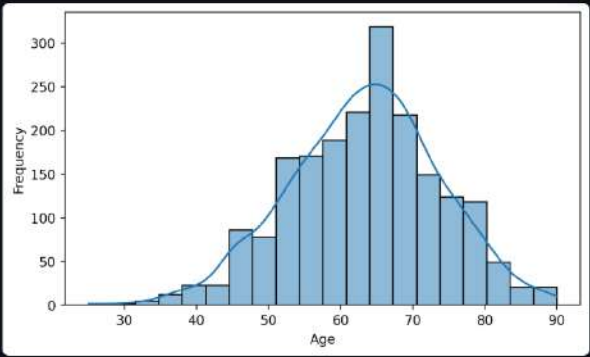
Model: Random Forest Classifier

Imbalance Handling: SMOTE

Risk Level Distribution



Age Distribution



Dataset Preview

	Patient_ID	Cancer_Type	Age	Gender	Smoking	Alcohol_Use	Obesity	Family_History	Diet_Red_Meat	Diet_Salted_Processed	Fruit_Veg_Intake	Physical_Activity	Air_Pollution	Occupational_Hazards	BRCA_M
0	LU0000	Breast	68	0	7	2	8	0	5	3	7	4	6	3	
1	LU0001	Prostate	74	1	8	9	8	0	0	3	7	1	3	3	
2	LU0002	Skin	55	1	7	10	7	0	3	3	4	1	8	10	
3	LU0003	Colon	61	0	6	2	2	0	6	2	4	6	4	8	
4	LU0004	Lung	67	1	10	7	4	0	6	3	10	9	10	9	

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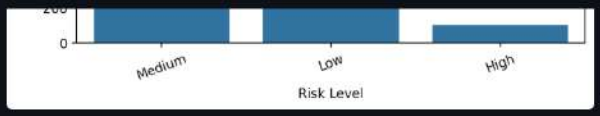
Navigation

Select a page

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Model: Random Forest Classifier

Imbalance Handling: SMOTE





Dataset Preview

	Patient_ID	Cancer_Type	Age	Gender	Smoking	Alcohol_Use	Obesity	Family_History	Diet_Red_Meat	Diet_Salted_Processed	Fruit_Veg_Intake	Physical_Activity	Air_Pollution	Occupational_Hazards	BRCA_M
0	LU0000	Breast	68	0	7	2	8	0	5	3	7	4	6	3	
1	LU0001	Prostate	74	1	8	9	8	0	0	3	7	1	3	3	
2	LU0002	Skin	55	1	7	10	7	0	3	3	4	1	8	10	
3	LU0003	Colon	61	0	6	2	2	0	6	2	4	6	4	8	
4	LU0004	Lung	67	1	10	7	4	0	6	3	10	9	10	9	
5	LU0005	Lung	77	1	10	8	3	0	6	0	6	2	10	7	
6	LU0006	Lung	59	0	10	10	0	0	9	4	0	1	10	9	
7	LU0007	Prostate	74	1	8	6	2	1	3	3	2	8	8	7	
8	LU0008	Colon	71	1	9	0	3	0	10	4	6	10	8	3	
9	LU0009	Skin	55	1	7	1	2	0	0	4	2	5	9	9	

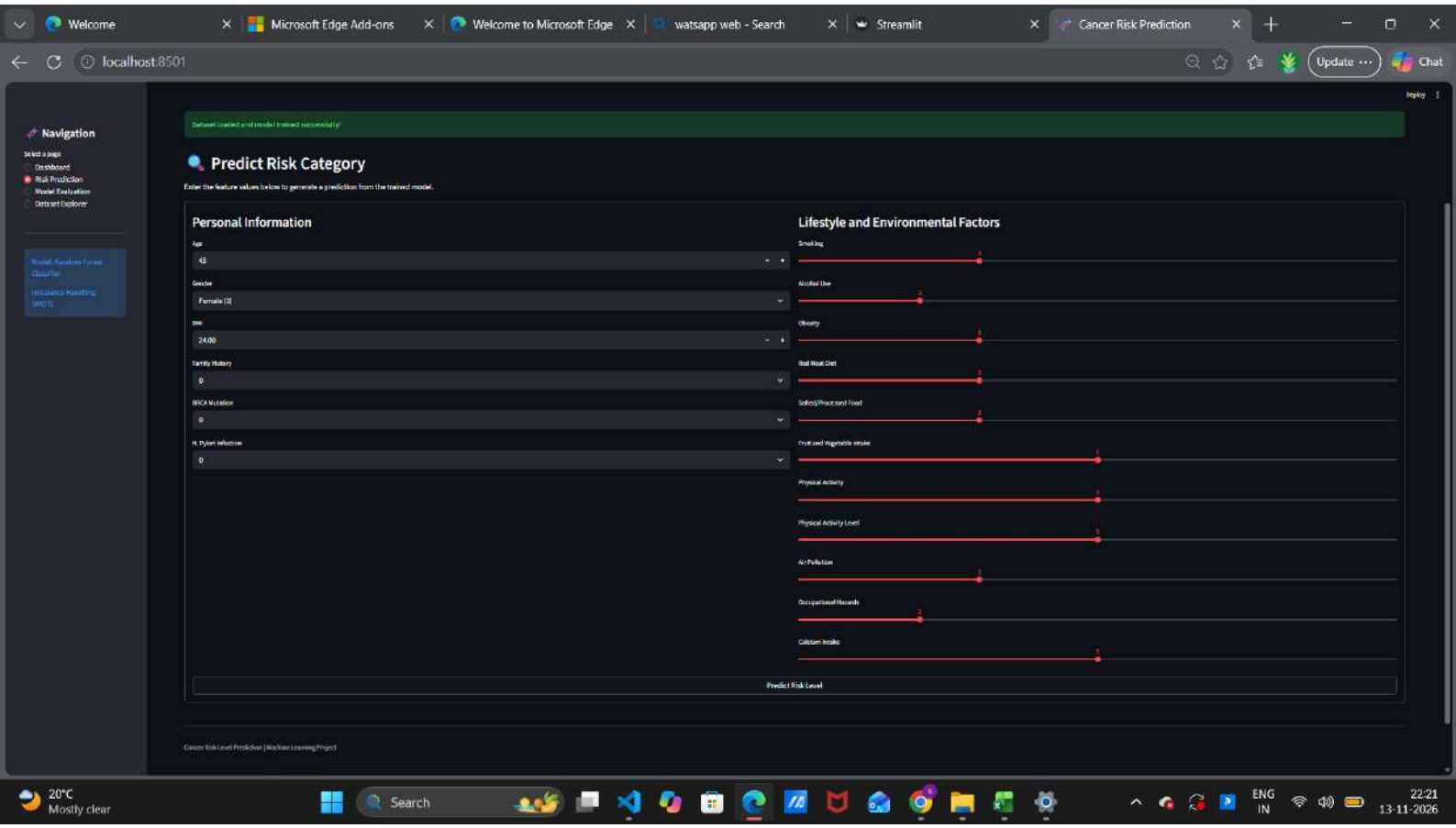
Cancer Risk Level Prediction | Machine Learning Project

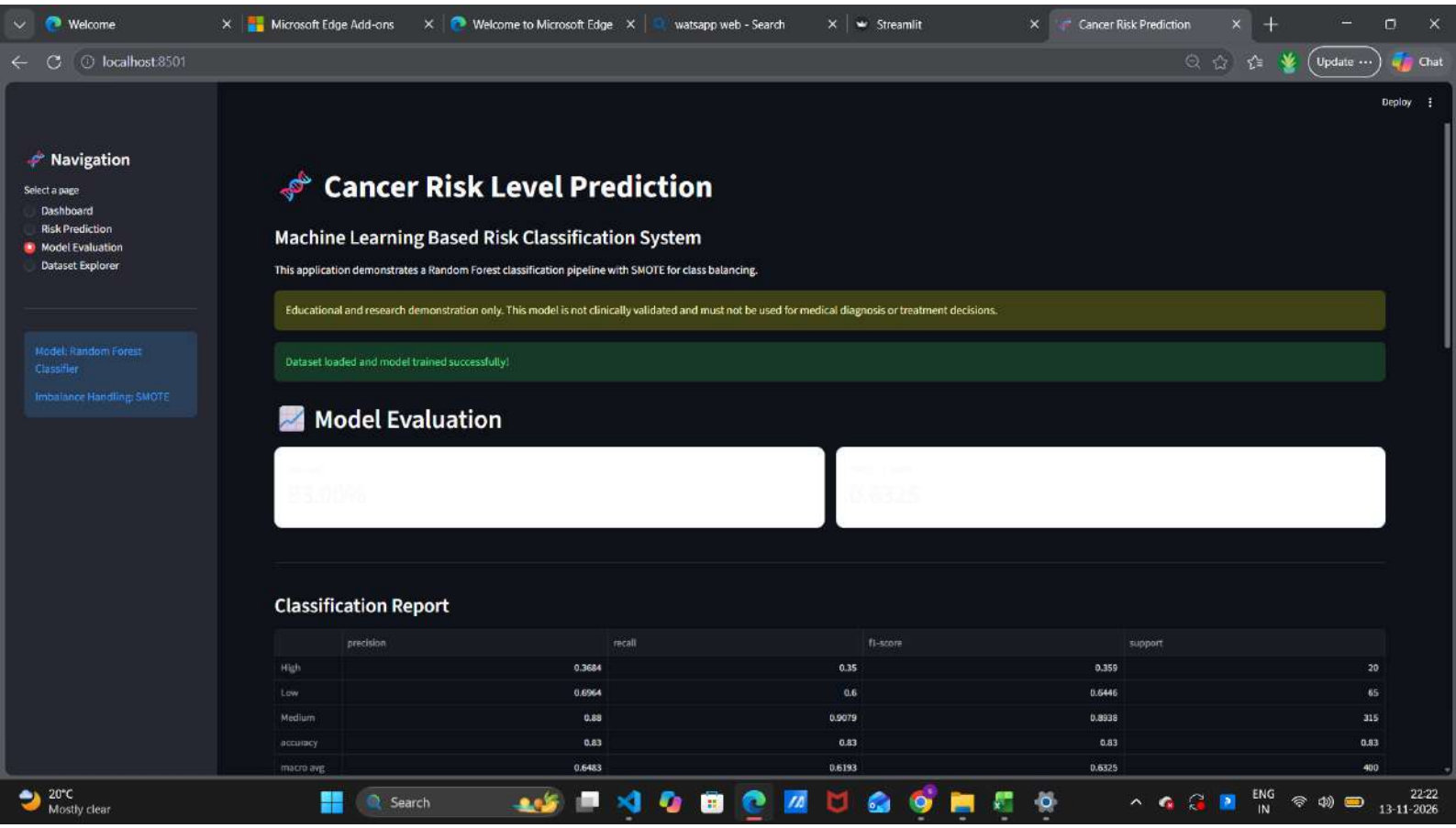
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Predict Risk Category
Enter the feature values below to generate a prediction from the trained model.
Personal Information
Age
Gender
BMI
Family History
BRCA Mutation
H. Pylori Infection
Lifestyle and Environmental Factors
Smoking
Alcohol Use
Obesity
Red Meat Diet
Salted/Processed Food
Fruit and Vegetable Intake
Physical Activity
Physical Activity Level
Air Pollution
Occupational Hazards
Calcium Intake
Predict Risk Level
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Dataset loaded and model trained successfully!
Model Evaluation
Classification Report
precision recall f1-score support
High 0.3684 0.35 0.359 20
Low 0.6964 0.6 0.6446 65
Medium 0.88 0.9079 0.8938 315
accuracy 0.83 0.83 0.83 0.83
macro avg 0.6483 0.6193 0.6325 400
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Model: Random Forest Classifier

Imbalance Handling: SMOTE

Classification Report

	precision	recall	f1-score	support
High	0.3684	0.35	0.359	20
Low	0.6964	0.6	0.6446	65
Medium	0.88	0.9079	0.8938	315
accuracy	0.83	0.83	0.83	0.83
macro avg	0.6483	0.6193	0.6325	400
weighted avg	0.8246	0.83	0.8265	400

Confusion Matrix



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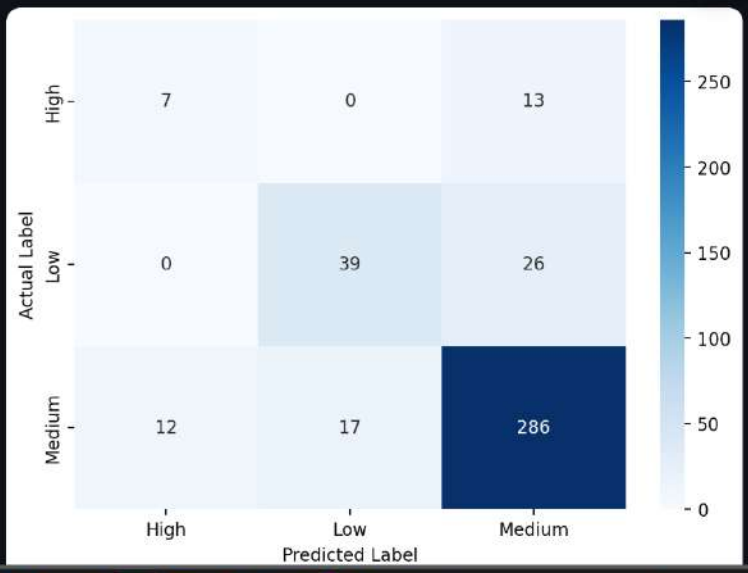
Navigation

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Model: Random Forest Classifier

Confusion Matrix

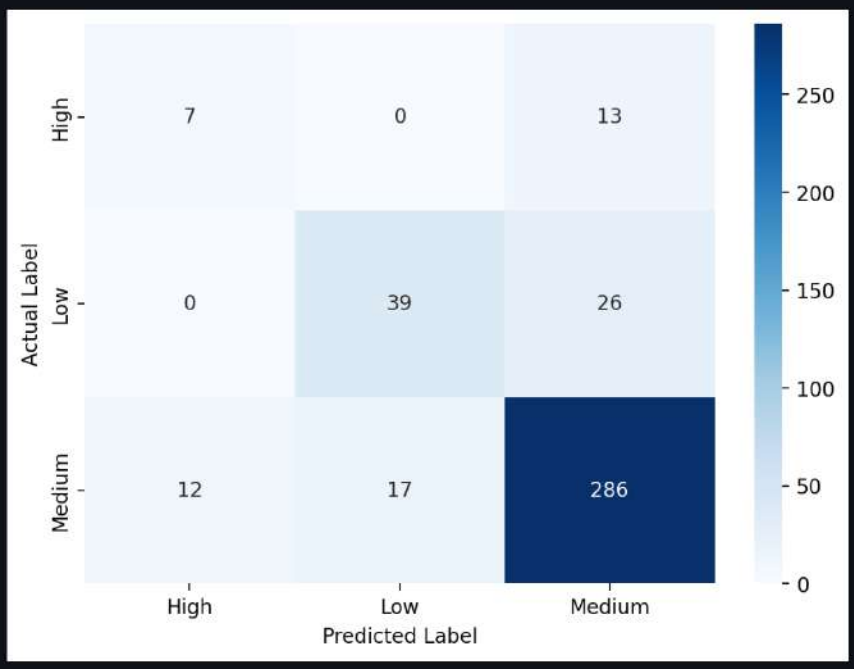


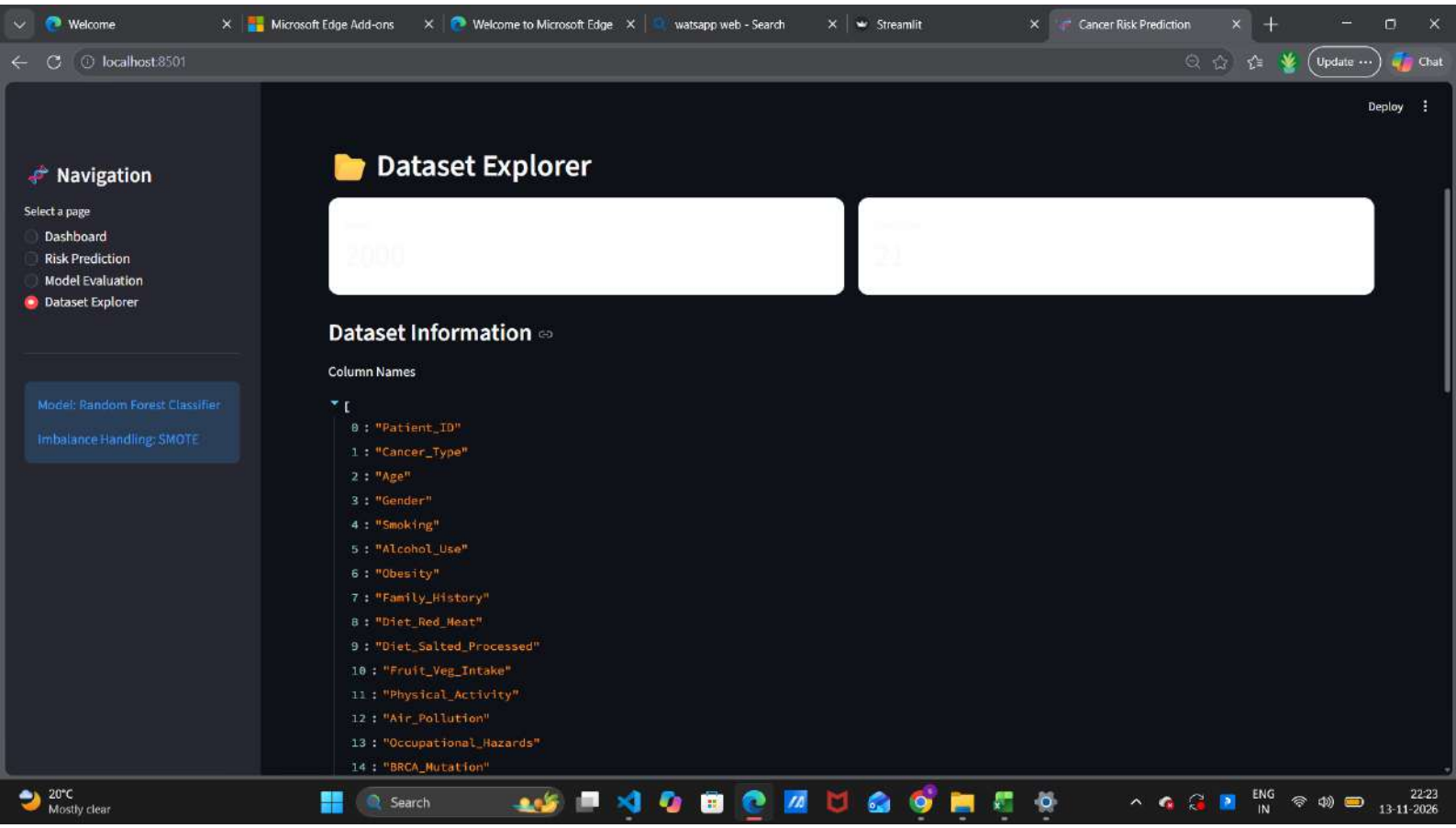
20°C Mostly clear

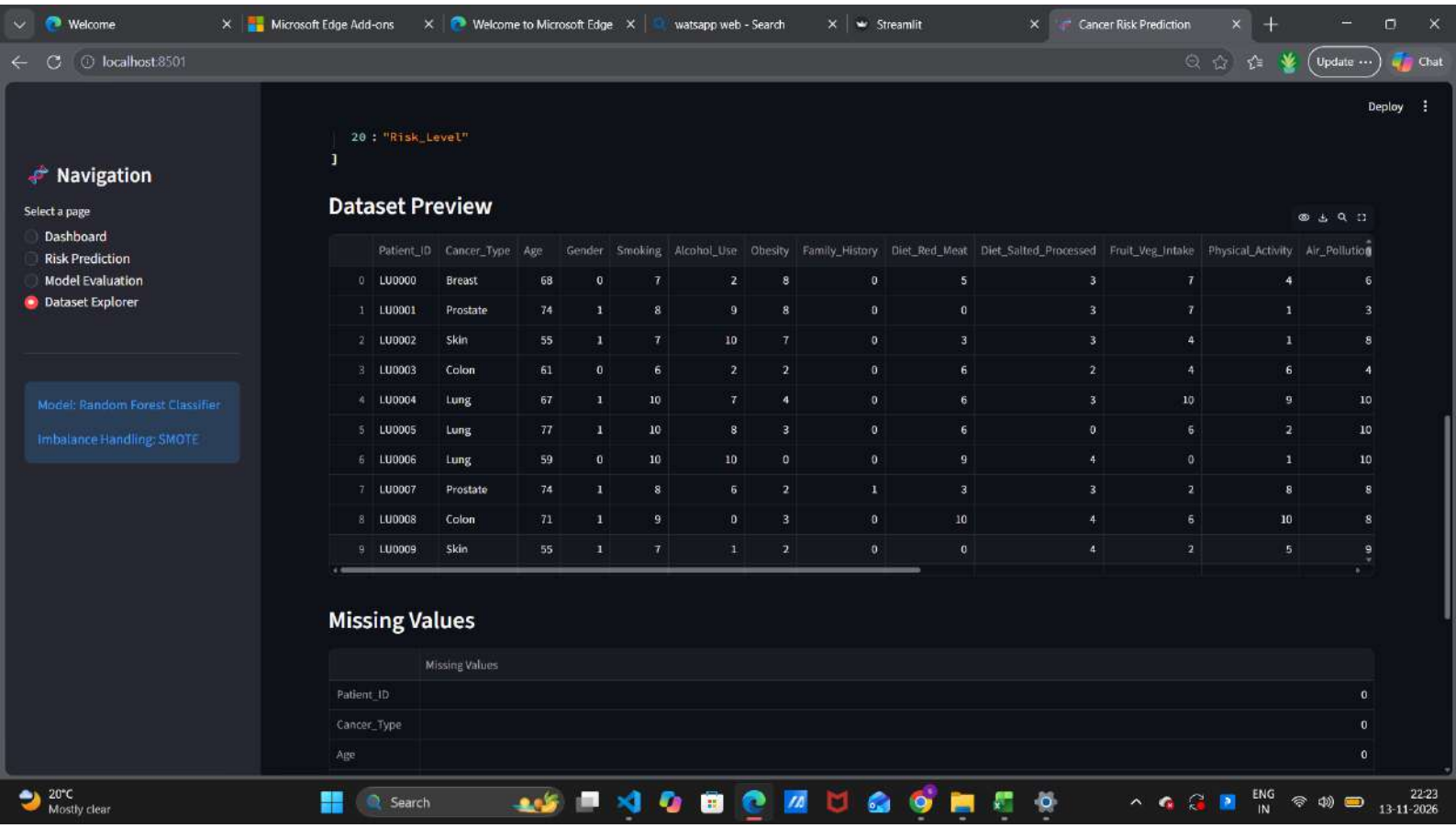
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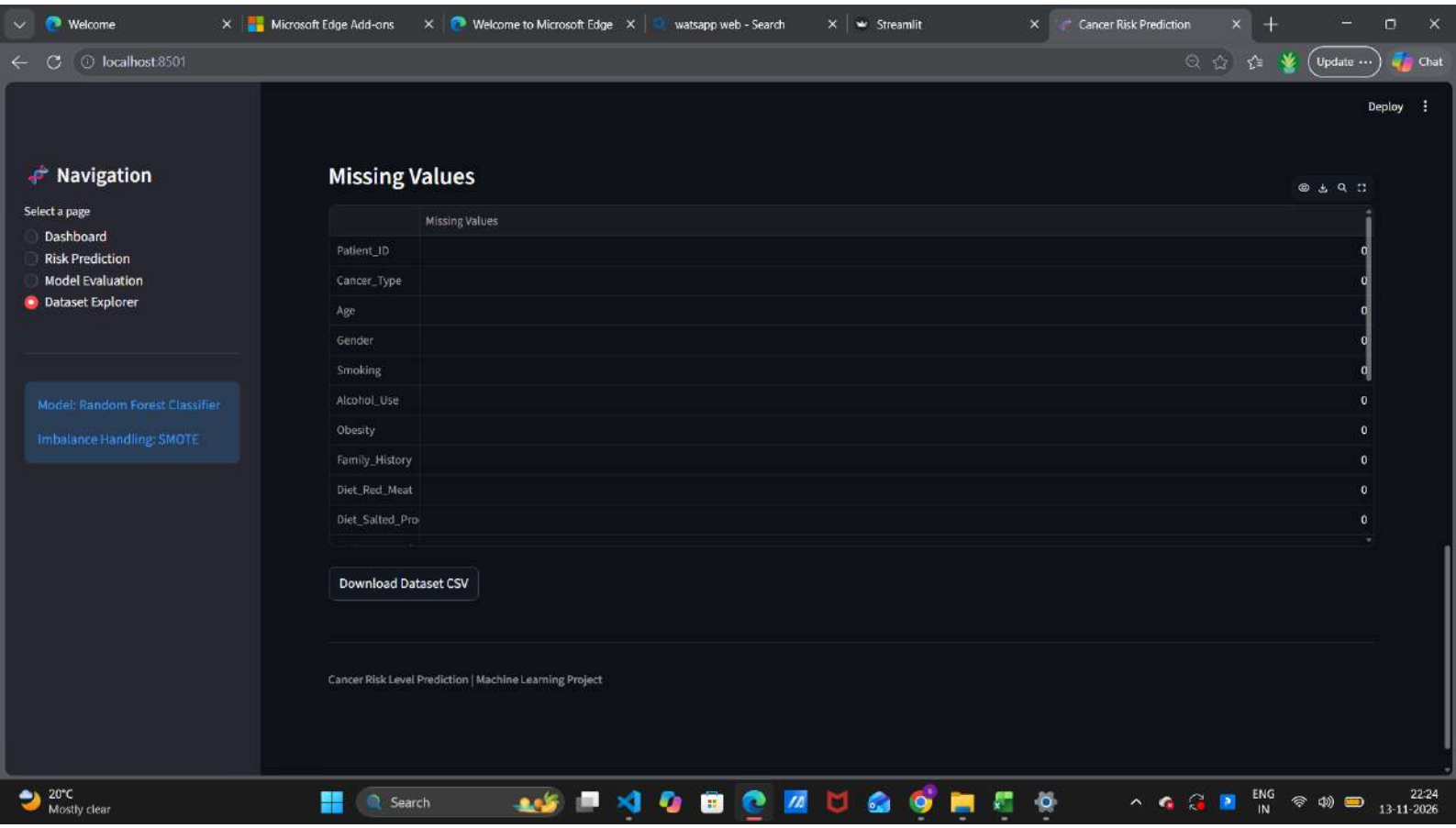
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Actual Label
High
Low
Medium
7
0
13
0
39
26
12
17
286
High
Low
Medium
Predicted Label
250
200
150
100
50
0


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Dataset Explorer
2000
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Dataset Information
Column Names
[
0 : "Patient_ID"
1 : "Cancer_Type"
2 : "Age"
3 : "Gender"
4 : "Smoking"
5 : "Alcohol_Use"
6 : "Obesity"
7 : "Family_History"
8 : "Diet_Red_Meat"
9 : "Diet_Salted_Processed"
10 : "Fruit_Veg_Intake"
11 : "Physical_Activity"
12 : "Air_Pollution"
13 : "Occupational_Hazards"
14 : "BRCA_Mutation"
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20 : "Risk_Level"
]
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Dataset Preview
Patient_ID Cancer_Type Age Gender Smoking Alcohol_Use Obesity Family_History Diet_Red_Meat Diet_Salted_Processed Fruit_Veg_Intake Physical_Activity Air_Pollution
0 LU0000 Breast 68 0 7 2 8 0 5 3 7 4 6
1 LU0001 Prostate 74 1 8 9 8 0 0 3 7 1 3
2 LU0002 Skin 55 1 7 10 7 0 3 3 4 1 8
3 LU0003 Colon 61 0 6 2 2 0 6 2 4 6 4
4 LU0004 Lung 67 1 10 7 4 0 6 3 10 9 10
5 LU0005 Lung 77 1 10 8 3 0 6 0 6 2 10
6 LU0006 Lung 59 0 10 10 0 0 9 4 0 1 10
7 LU0007 Prostate 74 1 8 6 2 1 3 3 2 8 8
8 LU0008 Colon 71 1 9 0 3 0 10 4 6 10 8
9 LU0009 Skin 55 1 7 1 2 0 0 4 2 5 9
Missing Values
Missing Values
Patient_ID 0
Cancer_Type 0
Age 0
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Missing Values
Missing Values
Patient_ID 0
Cancer_Type 0
Age 0
Gender 0
Smoking 0
Alcohol_Use 0
Obesity 0
Family_History 0
Diet_Red_Meat 0
Diet_Salted_Pro 0
Download Dataset CSV
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20°C
Mostly clear
Search
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IN
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